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Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Time: 03 Hours

Code: 105614 ***Subject: Web and Internet Technology*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven questions only): **[14]**

(a) Which of the following is NOT an Internet protocol?
(i) HTTP
(ii) SMTP
(iii) FTP
(iv) IDE

(b) A URL consists of
(i) Browser, client, port
(ii) IP address only
(iii) Protocol, domain name, path
(iv) HTML document only

(c) Which architecture involves multiple layers such as presentation, business logic, and database?
(i) Linear architecture
(ii) N-tier architecture
(iii) Client-side architecture
(iv) 1-tier architecture

(d) What does CSS stand for?
(i) Creative Style Sheet
(ii) Computer Style Sheet
(iii) Cascading Style Sheet
(iv) Color Styling Sheet

(e) Which HTML element is commonly used to create tables?
(i) <tlist>
(ii) <tbl>
(iii) <tab>
(iv) <table>

(f) Which of the following is NOT a feature of XML?
(i) XML is used to store data
(ii) XML displays data with rich formatting
(iii) XML is used to transport data
(iv) XML tags are user-defined

(g) A DTD is used to
(i) Format XML document
(ii) Create HTML style sheets
(iii) Define the structure and rules of an XML document
(iv) Validate JavaScript

(h) DOM stands for
(i) Document Object Model
(ii) Digital Online Model
(iii) Data Object Machine
(iv) Document Oriented Markup

(i) AJAX mainly works using
(i) PHP
(ii) Only HTML
(iii) XML + JavaScript + XMLHttpRequest
(iv) Only CSS

(j) PHP is a _____ scripting language.
(i) Client-side
(ii) Hardware-level
(iii) Server-side
(iv) Network-only

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Q.2 Solve both questions : **[14]**

(a) Describe the architecture of Client/Server computing. Compare fat client and fat server models with examples.

(b) What are markup languages? Discuss SGML, HTML, and XML along with their grammar and features.

Q.3 Solve both questions : **[14]**

(a) What is DHTML? Explain its features and compare HTML vs DHTML.

(b) Describe the Browser Object Model (BOM) and its role in dynamic web page management.

Q.4 Solve both questions : **[14]**

(a) Explain the importance and features of XML. Describe the structure of an XML document with an example.

(b) What is a Document Type Definition (DTD)? Explain internal and external DTD with examples.

Q.5 Solve both questions : **[14]**

(a) Explain XML Schema, its structure, data types, and importance compared to DTD.

(b) Describe the main characteristics of JavaScript and explain how JavaScript is used to create dynamic HTML pages.

Q.6 Solve both questions : **[14]**

(a) Explain the architecture and components of AJAX. How does AJAX help in dynamic web page updating?

(b) Describe the structure and working of the XMLHttpRequest object.

Q.7 Solve both questions : **[14]**

(a) Discuss how XML can be accessed, created, modified, and loaded using AJAX.

(b) Explain the process for handling response XML in an AJAX-based application.

Q.8 Solve both questions : **[14]**

(a) Explain the structure of PHP and discuss its major features for web application development.

(b) Describe how AJAX interacts with PHP in a web application. Explain the complete flow with a suitable example.

Q.9 Solve both questions : **[14]**

(a) Explain the steps involved in connecting an ASP application to a database.

(b) Write notes on ASP objects, variables, control structures, and scripting components.

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